

Fall 2025 05-332/632: Introduction to Accessibilind Assistive Technology

Meeting Time

Classroom:

GHC 4215

Day/Time:

Mondays and Wednesdays

11:00 am - 12:20 pm

Links

Website: cmu-05-332.github.io

Canvas: <https://canvas.cmu.edu/courses/47948>



Course Description and Learning Goals

This course will explore ability, disability, and accessibility. It introduces students to the theory, history, policy, and practice of accessibility and inclusion in computing. Course readings will explore relevant concepts from disability studies, accessibility, human-computer interaction, and design methods. Through this course, you will learn how to design for people with different abilities including blind or low vision, hearing loss, motor impairments, cognitive decline, and neurodiversity. As early adopters, people with disabilities have inspired a host of future user interface technologies,

e.g., conversational assistants, text-to-speech, speech recognition, optical character recognition, predictive typing, tactile displays, etc. People with disabilities continue to be the first users of next-generation interface technologies that are gradually adopted more widely including Gen-AI and image recognition tools. We will discuss how to design online and offline; how to design for visible and invisible disabilities; and how to design for permanent, situational, and temporary disabilities.

Students will leave the course with a sense of:

- What access technologies already exist and what technologies are emerging
- Critique different models of disability based on literature in disability studies
- Define and distinguish between web accessibility guidelines and legislation
- What it takes to both create and use accessible computing systems for people with visible and invisible disabilities
- How to make interfaces and interactions more accessible
- How to make documents accessible and conduct inclusive presentations and meetings
- Critically discuss accessibility approaches of different media formats across disabilities

Course Staff and Logistics

Instructors



Patrick Carrington

Associate Professor, HCII

Office Location: 407 S. Craig Street, Room 201

Office hours:

Office hours will be on Mondays and Wednesdays.

Please sign up through Calendly at least 24 hours in advance.

If you need to meet outside of these hours email Dr. Carrington.



Franklin Mingzhe Li

Ph.D. Candidate, HCII

Office Location: 407 S. Craig Street, Room 212

Office hours:

Office hours will be on Tuesday and Thursday. Please email Franklin to schedule available time.

Waitlist

We will add students from the waitlist as seats become available. Waitlists are processed in FIFO order. If you are on the waitlist, we recommend attending the first couple of weeks of class and attempting the first homework.

Pre-Requisites

There are no prerequisites for this course. However, some prior programming experience or knowledge of web programming is helpful but not required.

Communication

We will have a Slack group for this class to help facilitate timely communication and coordination. We will make announcements through a dedicated channel in Slack, and also provide channels for homework clarifications and questions relevant to everyone in the class. As such, we highly recommend that students install Slack on their phones / desktop with notifications enabled.

Questions about individual grades, personal questions, or any other potentially confidential matters should be submitted via email.

You can contact the course instructors either through Slack or email.

Readings

Various readings throughout the semester will be made available online or through the library; we do not have a single textbook but rather assemble readings from different sources. Readings will be posted to the course website and/or made available via Canvas.

Assessment

Course Element	Grade Percentage
Homework/Mini-Projects	55%
Reading Reflection	5%
In-Class Activities	10%
Final Project	30%

Late Work and Regrade Policy

Late assignments are accepted however you may not receive the same level of feedback. This will be especially true in the case of peer-reviewed assignments as you may not be able to participate and thus will not receive the same level of feedback. In most cases, late homework assignments **may be docked 10% for each day that it is late.** The maximum late penalty is 50% of the total assignment grade.

Due to heavy emphasis on planning & teamwork in this course **there are no late days for project deliverables;** We expect that you anticipate unexpected events in your planning, and coordinate with your teammates when they arise. Exceptions to this policy will be made only in extraordinary circumstances.

If you have an issue or a question about your grade and would like to request a **regrade** of that assignment, these must be requested within 1 week after grades have been released for a particular assignment. Regrade requests must be submitted via email to both instructors.

Participation

Class attendance and participation are important parts of the learning in this course. To account for this, a portion of the final grade is based on your regular attendance and active participation (see Assessment section). That said, we also recognize that students may need to miss class for a variety of reasons (religious observance, job interview, university-sanctioned event, or illness). For that reason, **all students are permitted two class absences for any reason** without any impact on the final grade. This includes both excused or unexcused absences. When you must miss class, please notify the instructors (at least 24 hours in advance except for illness/emergency), so that we can discuss alternative arrangements for catching up on class and associated work. If you encounter extenuating circumstances and must miss more than two classes, please come and discuss the issue with the instructors; we would like to find a way to support you.

If you find yourself in a situation that calls for an extended absence for other reasons, please have your academic advisor reach out to the instructors, and they will work to find a solution on a case by case basis.

Time Management

This is a 12-unit course, and it is our intention to manage it so that you spend close to 12 hours a week on the course, on average. In general, ~3 hours/week will be spent in lectures and 9 hours on readings, homework, and projects.

Throughout the semester, please feel free to give the course staff feedback on how much time the course is taking for you.

Writing

Describing tradeoffs among decisions and communication with less technical stakeholders are important skills to develop. Most projects have a component that requires discussing issues in written form or reflecting about experiences. To practice writing skills, the Global Communications Center (GCC) offers one-on-one help for students, along with workshops. The instructors are also happy to provide additional guidance if requested.

Policy on use of Generative AI for Writing

You may use generative AI technologies such as ChatGPT or CoPilot for assisting in code development or client communications, unless the question/assignment specifically prohibits it. However, any use of AI comes with two caveats: (1) you must clearly indicate the use of such technology (and name the tool) in every homework question or source code file where an AI tool was used as a form of attribution, just as you would if you had taken help from a friend; (2) you are liable for factually inaccurate answers or unspecific rambling produced by AI tools; it is your responsibility to edit AI-produced content before submitting it for class purposes.

Recordings

For this course, we will be recording most class sessions and making them available to you for your personal, educational use. Recordings of class sessions are covered under the Family Educational Rights and Privacy Act (FERPA) and must not be shared with anyone outside your course-section. The purpose of these recordings is so students in this course (and only students in this course) can watch or re-watch past class sessions. Feel free to use the recordings if you would like to review something we discussed in class or if you are temporarily unable to attend class.

Devices in Lecture

Research shows that using devices on non-class related activities harms both the device user's learning, and other students' learning as well. Therefore, in general, we do not allow the use of any devices during lecture unless required for a specific in-class activity.

Academic Honesty

The usual policies apply, especially the University Policy on Academic Integrity. If any of your work contains a statement that was copied verbatim from an external source, you must put it in quotes and cite the source. If you are paraphrasing an idea you read elsewhere, you must acknowledge the

source. If you are using a tool to help format or structure text significantly based on content and facts that you provided, you must acknowledge the use of such tools (see policy on generative AI tools above). Using existing material without proper citation is plagiarism, a form of cheating. If there is any question about whether the material is permitted, you must get permission in advance. We will be using automated systems to detect software plagiarism.

Any violation of this policy is cheating. The minimum penalty for cheating (including plagiarism) will be a zero grade for the whole assignment. Cheating incidents will also be reported through University channels, with possible additional disciplinary action (see the above-linked University Policy on Academic Integrity).

If you have any question about how this policy applies in a particular situation, ask the instructors for clarification.

Note that the instructors respect honesty in these (and indeed most!) situations.

A Note on Self Care

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep and taking some time to relax. This will help you achieve your goals and cope with stress. Please understand that you can reach out if you do need help with something.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit <http://www.cmu.edu/counseling/>. Consider reaching out to a friend, faculty or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

- CaPS: 412-268-2922
- Re:solve Crisis Network: 888-796-8226
- If the situation is life threatening, call the police:
 - On campus: CMU Police: 412-268-2323
 - Off campus: 911

Food Insecurity

If you are worried about affording food or feeling insecure about food, there are resources on campus that can help. Any undergraduate or graduate student can visit the CMU Pantry and receive food for free. Follow the directions on the CMU Pantry website to schedule your visit.